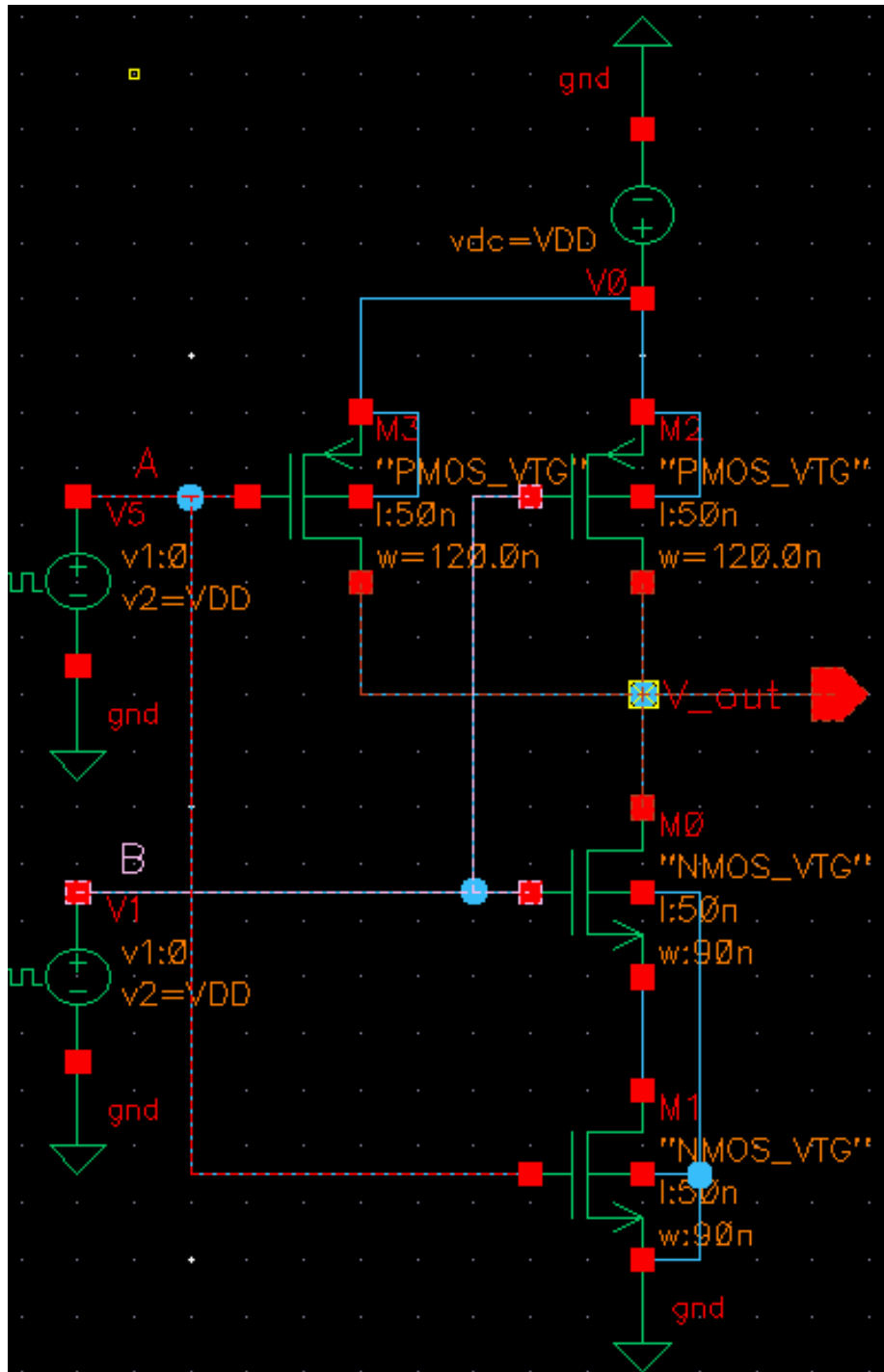


CSE 112 Project

David Kim

Schematic:



Waveforms:

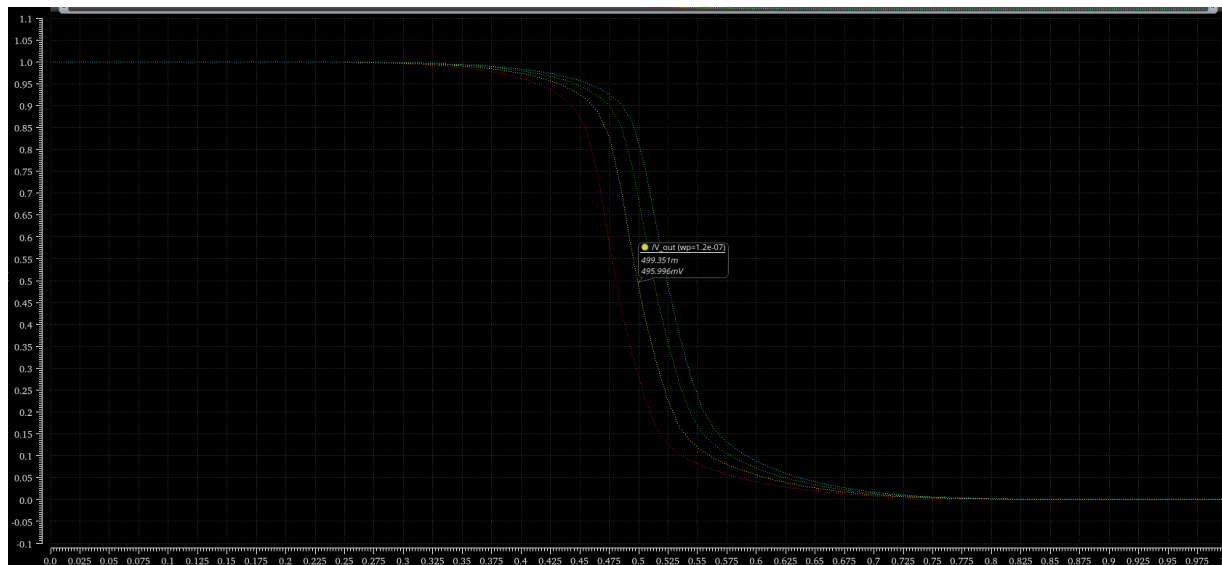
https://docs.google.com/spreadsheets/d/1bFX0t1NfLq-Lu4ImUwS3jrNgEhvjEsX2_uCLdzvJLRY/edit?gid=1863969740#gid=1863969740

The red line is the voltage of input A. The voltage is high for the first nanosecond and low for the second nanosecond.

The blue line is the voltage of input B. The voltage alternates every half nanosecond, starting at a high voltage for the first half nanosecond.

The yellow line is the output voltage. The output voltage is low for the first half nanosecond, when both signals are high, and high for the rest of the waveform. There is a slight dip at around the 1 nanosecond mark when input A transitions from high to low, but it is short and only for the duration of input A's transition.

Optimization:



<https://drive.google.com/file/d/1X07Y80OH-m3jWk9zvbrcy7glwIx5eCcZ/view?usp=sharing>

The red line is the VTC for $W_p=90$ nm.

The yellow line is the VTC for $W_p=120$ nm.

The green line is the VTC for $W_p=150$ nm.

The blue line is the VTC for $W_p=180$ nm.

For $V_{DD} = 1$, the yellow line is the closest to crossing through the point (0.5, 0.5), meaning that the VTC for $W_p = 120$ nm is closest to being symmetrical.

Therefore, the optimal $W_p/W_n = 120\text{nm}/90\text{nm} = 1.333$.

For the rest of this report, the transistors were sized according to this ratio.

Delay Analysis:

Delay vs Supply Voltage:

https://docs.google.com/spreadsheets/d/1bFX0t1NfLq-Lu4ImUwS3jrNgEhvjEsX2_uCLdzvJLRY/edit?gid=2064949860#gid=2064949860

As the supply voltage increases from 0.8 to 1.2 volts, the propagation delay decreases at a decreasing rate.

Delay vs temperature:

https://docs.google.com/spreadsheets/d/1bFX0t1NfLq-Lu4ImUwS3jrNgEhvjEsX2_uCLdzvJLRY/edit?gid=1510416897#gid=1510416897

As the temperature increases from 0 to 90 degrees, the propagation delay increases at a relatively linear rate.

Conclusions:

When designing a NAND gate, or any gate, there is always a tradeoff for any choice. One such tradeoff is found in sizing the NAND gate's transistors. A larger width takes up more space and results in a greater capacitance, which increases delay. Conversely, a smaller width results in less current, which also increases delay, while also making the gate more susceptible to noise, resulting in worse robustness. Sizing also affects the symmetry of the NAND gate, as we found in the optimization section above. A larger supply voltage could result in increased power consumption, but it also reduces delay, as we found in the Delay Analysis section. The effectiveness of digital logic gates are determined by many factors such as physical space, power, speed, and robustness. These factors make perfect optimization of logic gates impossible as there are tradeoffs when trying to optimize one of these factors. Therefore, gate designers must keep in mind these tradeoffs and how performance might be affected.